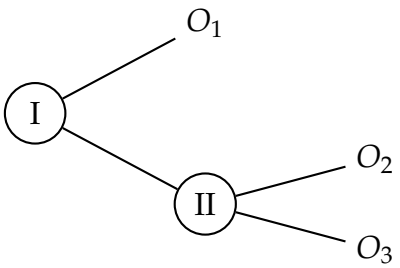
MTH636/SDS636 (2026-27, ODD SEMESTER)

PRACTICE PROBLEM SET 3

1. Depict the following situation as a game in extensive form. Peter and his three children, Andrew, James, and John, manage a communications corporation. Andrew is the eldest child, James the second-born, and John the youngest of the children. Two candidates have submitted offers for the position of corporate accountant at the communications corporation. The choice of a new accountant is conducted as follows: Peter first chooses two of his three children. The two selected children conduct a meeting to discuss the strengths and weaknesses of each of the two candidates. The elder of the two children then proposes a candidate. The younger of the two children expresses either agreement or disagreement with the proposed candidate. A candidate is accepted to the position only if two children support his candidacy. If neither candidate enjoys the support of two children, both candidates are rejected.

2. Consider the game in the following figure



where O_1, O_2, and O_3 are distinct and taken from the set $\{(1,-1),(-1,1),(0,0)\}$.

   (a) Is there a choice of O_1, O_2, and O_3 such that Player I can guarantee victory for himself? Justify your answer.

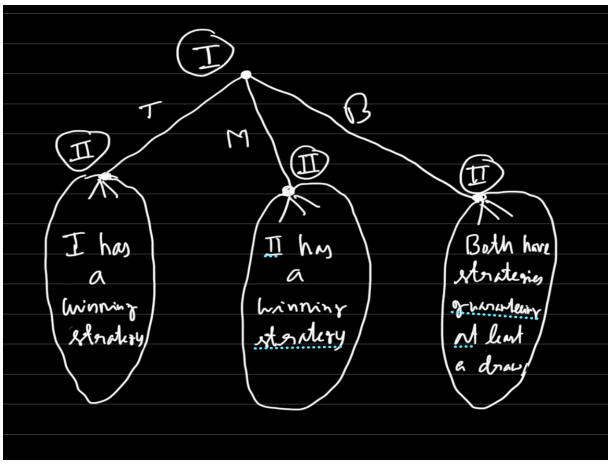


Figure 1

(b) Is there a choice of O_1, O_2, and O_3 such that Player II can guarantee victory for himself? Justify your answer.

(c) Is there a choice of O_1, O_2, and O_3 such that both players can guarantee for themselves at least a draw? Justify your answer.

3. Recall the CHOMP game by David Gale. Answer the following questions.

   (a) Which of the two players has a winning strategy in a game of Chomp played on a $2 \times \infty$ board? Justify your answer. Describe the winning strategy. (Recall that the proof we did in class holds under the assumption that the board is finite.)

   (b) Which of the two players has a winning strategy in a game of Chomp played on an $m \times \infty$ board, where m is any finite integer? Justify your answer. Describe the winning strategy.

4. Consider the following game with perfect information and without chance moves, where for each of the subgames the alternative that holds (from von Neumann's theorem) is written within the subgame (see Figure 1). Find which alternative according to von Neumann's theorem is true for this game. Also, find the strategy/strategies?

5. Consider the following game (known as the Nim game in the literature). It is a game between two-player where two piles of matches are placed before the players, and each pile has three matches. Each player in turn chooses a pile and removes any number of matches from the pile he has selected (he must remove at least one match). The player who removes

the last match wins the game. Note that by von Neumann's Theorem (Theorem 1 in the slides), one of the players has a winning strategy in this game.

(a) Find who has a winning strategy and find the winning strategy.

(b) What will happen in the general case, i.e., when the two plies have x and y number of matches for some $x, y \in \mathbb{N}$.

6. Establish whether there exists a two-player game in extensive form with perfect information, and possible outcomes I (Player I wins), II (Player II wins), and D (a draw), whose strategic-form description is

		Player II			
		s_{II}^1	s_{II}^2	s_{II}^3	s_{II}^4
Player I	s_{I}^1	D	I	II	I
	s_{I}^2	I	II	I	D
	s_{I}^3	I	I	II	II

If the answer is yes, describe the game. If not, explain why not.

7. A pack of three cards, labelled 1, 2, and 3, is shuffled. Sachin, Sourav, and Rahul each take a card from the pack. Each of the two players holding a card with low values (1 or 2) pays the amount of money appearing on the card he or she is holding to the player holding the high-valued card (namely, 3). Depict this situation as a game in extensive form.

8. Establish whether there exists a two-player game in extensive form with perfect information, and possible outcomes I (Player I wins), II (Player II wins), and D (a draw), whose strategic-form description is

		Player II			
		s_{II}^1	s_{II}^2	s_{II}^3	s_{II}^4
Player I	s_{I}^1	D	D	D	D
	s_{I}^2	I	I	II	II
	s_{I}^3	II	II	D	D

If the answer is yes, describe the game. If not, explain why not.

9. Consider the following two-player game. Initially there are n matches on a table. The players move alternately, and on each turn a player must remove either 1, 2, or 3 matches. The player who removes the last match wins.

   (a) Determine which player has a winning strategy when $n = 10$, and describe a winning strategy.

   (b) Characterize all values of n for which Player I has a winning strategy.

   (c) Now suppose that on each turn a player may remove either 1, 3, or 4 matches. Characterize the winning and losing positions.

10. Consider a game of Nim with three piles containing 1, 2, and 3 matches, respectively. As usual, on each turn a player chooses exactly one pile and removes any positive number of matches from that pile. The player removing the last match wins.

    (a) Which player has a winning strategy? Describe one.

    (b) Answer the same question when the initial pile sizes are 2, 3, and 4.

    (c) Without using the general theorem for Nim, identify all losing positions (x, y, z) with

    $$0 \leq x \leq y \leq z \leq 3.$$

11. There are $2n$ points equally spaced on a circle. Two players alternately draw a straight line segment joining two previously unjoined points. A move is legal only if the new segment does not intersect any previously drawn segment except possibly at its endpoints. A player who has no legal move loses.

    Find out which player has a winning strategy and find one such winning strategy?

12. A rectangular chocolate bar consists of $m \times n$ squares. Two players alternately make one straight cut along the lines separating the squares. A cut must divide exactly one existing piece into two pieces; pieces may not be stacked. The player who makes the last possible cut wins.

(a) Determine which player has a winning strategy for a 3×4 bar.

(b) Determine which player has a winning strategy for an arbitrary $m \times n$ bar.

(c) Does the answer depend on the choices made by the players during the game? Explain.